AquaGuard

Real-Time Floating Garbage Detection and Pollution Severity Assessment in Sri Lankan Rivers Using YOLOv8 Deep Learning with Weighted Garbage Density Index

Research Overview Document

Literature Review | Research Gap Analysis | System Architecture
Dataset Strategy | Methodology | Implementation Plan

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1. Research Title and Objective

Proposed Title: Real-Time Floating Garbage Detection and Pollution Severity Assessment in Sri Lankan Rivers Using YOLOv8 Deep Learning with Weighted Garbage Density Index

Research Objective: To develop an end-to-end system that detects floating garbage in Sri Lankan rivers using YOLOv8 object detection, classifies waste into four categories (plastic, polythene, glass, metal), calculates a weighted Garbage Density Index (GDI) for quantifying pollution severity, and provides location-based pollution tracking with automated cleanup prioritization through a web-based dashboard.

Problem Statement: Sri Lanka has 103 rivers, and the Kelani River is identified as the most polluted. Over 7,000 factories discharge waste into the Kelani river valley. Despite recent efforts like the NARA debris boom deployment in the Ja-Ela River, there is no automated visual monitoring system for quantifying river garbage pollution. Existing research focuses solely on detection accuracy but provides no mechanism to convert detections into environmental management decisions.

2. Literature Review — 5 Key Research Papers

Paper 1: Enhanced YOLOv8 for Floating Object Detection on Water Surfaces

Cao et al. (2025) | Scientific Reports (Nature) | DOI: 10.1038/s41598-025-32817-x

Aspect	Details
Algorithm	SEDS-YOLOv8 (improved YOLOv8-nano)
Problem	Water reflections, ripple noise, and small debris confuse standard detectors
Key Techniques	SE attention (suppresses reflection noise) + DSConv (7.3% more efficient) + EIoU loss (better box accuracy)
Dataset	Hybrid: 3,000 ORCA-UBOAT images + 2,000 web-scraped images = 28,000 after augmentation
Results	mAP@0.5: 88.82%, Precision: 86.02%, Recall: 85.01%, Speed: 103.7 FPS
End System	Detection only - no pollution scoring, no dashboard, no deployment system
Limitation	No garbage type classification, no pollution measurement, no real-world application

Paper 2: eCBAM-YOLOv11 for Riverine Floating Garbage Classification

Zhao et al. (2026) | Applied Sciences (MDPI) | DOI: 10.3390/app16020651

Aspect	Details
Algorithm	Modified YOLOv11 with eCBAM attention + saSIoU loss
Problem	Multi-size garbage, irregular shapes, complex water backgrounds
Key Techniques	Enhanced CBAM (multi-scale channel + boundary spatial attention) + Scenario-adapted SIoU loss
Dataset	Self-collected: 12,000 images, 6 garbage categories from real rivers
Results	mAP@0.5: 86.48%, Precision: 80.43%, Recall: 84.36%
End System	Detection only - no web application, no pollution index
Limitation	No pollution severity scoring, no temporal tracking, no deployment framework

Paper 3: MFPF-YOLOv11s for Water-Floating Garbage Detection

Wang et al. (2026) | Frontiers in Computer Science | DOI: 10.3389/fcomp.2026.1859574

Aspect	Details

Algorithm	MFPF-YOLOv11s (modified YOLOv11-small)
Problem	Stacked/occluded garbage, water surface fluctuations, illumination variation
Key Techniques	Multi-stage dilated residual module + Meta-ACON activation + Context Aggregation attention
Dataset	Yellow River water-floating garbage dataset (custom collected)
Results	5-6% improvement over baseline YOLOv11 in mAP metrics
End System	Detection only - academic model comparison
Limitation	No end-to-end system, no garbage type weighting, no location tracking

Paper 4: YOLOv8-SST for Small Floating Objects in Complex Water Environments

Dong et al. (2026) | PLOS ONE | DOI: 10.1371/journal.pone.0340822

Aspect	Details
Algorithm	YOLOv8-SST (Small Surface Targets)
Problem	Small floating objects missed by standard detectors
Key Techniques	C2fBF module (BiFormer dual-layer routing attention) + 160x160 small target detection layer
Dataset	Public water surface datasets with augmentation
Results	Significant improvement for small targets over baseline YOLOv8
End System	Detection only - no garbage type classification (only generic "floating object")
Limitation	Cannot distinguish plastic from metal - detects only generic objects, no severity measurement

Paper 5: Image Analysis of River-Floating Waste Using Deep Learning

2024 | MDPI Water | DOI: 10.3390/w16101373

Aspect	Details
Algorithm	YOLOv5 + DeepSORT (detection + tracking)
Problem	Counting and tracking waste flowing in rivers over time
Key Techniques	Object tracking with unique IDs - prevents double-counting as garbage floats downstream
Dataset	Laboratory flume experiments with 7 waste classes (cans, cartons, plastic bottles, foams, glasses, papers, plastics)
Results	Good in clear conditions, accuracy drops for submerged/clumped garbage
End System	Lab-only setup - not deployed in real rivers
Limitation	Lab experiments only, no real-river deployment, no pollution measurement system

Sri Lanka Context Paper: ML for Kelani River Pollution Sources

Wijayaweera et al. (2024) | MDPI Water | DOI: 10.3390/w16192766

This Sri Lankan paper used chemical water quality sensor data (not computer vision) to identify industrial pollution sources in the Kelani River Basin. Key finding: 'The Kelani River Basin covering only ~3.6% of Sri Lanka's land but hosting over a quarter of its population is identified as the most polluted watershed in the country.' **No one has used computer vision for garbage detection on Sri Lankan rivers.**

3. Comparative Analysis — AquaGuard vs Existing Systems

The table below compares what each paper's system can do versus what AquaGuard provides. Green cells indicate the feature is present; red indicates it is absent.

Feature	Paper 1 Cao 2025	Paper 2 Zhao 2026	Paper 3 Wang 2026	Paper 4 Dong 2026	Paper 5 2024	AquaGuard (Ours)
Garbage detection	Yes	Yes	Yes	Yes	Yes	Yes
Multi-class classification	No	Yes (6)	No	No	Yes (7)	Yes (4)
Pollution severity score	No	No	No	No	No	YES - GDI
GPS location tracking	No	No	No	No	No	YES
Temporal trend analysis	No	No	No	No	No	YES
Web dashboard	No	No	No	No	No	YES
Mobile camera support	No	No	No	No	No	YES
Alert system	No	No	No	No	No	YES
Bounding box visualization	Yes	Yes	Yes	Yes	Yes	YES
Real-river deployment	No	No	No	No	No	YES - Sri Lanka
Cleanup prioritization	No	No	No	No	No	YES
REST API backend	No	No	No	No	No	YES

Key Finding: All 5 papers stop at detection. AquaGuard is the only system that provides a complete pipeline from detection to actionable pollution management. This is the fundamental research gap that AquaGuard addresses.

4. Research Gap and Novel Contributions

4.1 Identified Research Gap

After systematic review of all five papers, a clear pattern emerges: every existing work treats garbage detection as the end goal. They improve mAP scores, precision, and recall, but provide no mechanism to convert detections into environmental management decisions. A model that draws bounding boxes on images, however accurately, gives no actionable information to cleanup authorities. The gap is not in detection — it is in what happens AFTER detection.

4.2 Three Novel Contributions

Novel Contribution 1: Weighted Garbage Density Index (GDI)

No existing paper calculates a weighted pollution score based on garbage type. AquaGuard assigns danger weights: polythene (1.0) > plastic (0.8) > glass (0.7) > metal (0.6), because polythene bags are more environmentally harmful in rivers (they block water flow, don't decompose, suffocate aquatic life). The GDI formula: GDI = Sum(confidence x danger_weight) / area_normalisation x scale. This converts raw detection counts into an actionable 0-100 pollution score classified as Low (0-25), Moderate (25-50), High (50-75), or Critical (75-100).

Novel Contribution 2: Location-Based Temporal Pollution Tracking

AquaGuard stores GPS coordinates + GDI score + timestamp for every scan, enabling: pollution heatmaps showing which river sections are worst, trend analysis revealing whether pollution is increasing or decreasing, cleanup prioritization directing limited resources to the most polluted areas first, and before/after measurement proving whether cleanup efforts are effective. No existing paper tracks pollution across locations or over time.

Novel Contribution 3: First Visual Garbage Detection for Sri Lankan Rivers

The only Sri Lankan paper (Wijayaweera et al., 2024) used chemical water quality sensors, not computer vision. AquaGuard is the first system to apply deep learning-based visual garbage detection to Sri Lankan river environments. The custom dataset captures local garbage types (Lanka-specific packaging), local water conditions (monsoon turbidity), and local vegetation patterns that global models cannot account for.

5. System Architecture

5.1 System Pipeline — 7-Layer Architecture

The AquaGuard system operates as a 7-layer pipeline where each layer feeds into the next:

Layer	Component	Technology	Function
1	Input	Camera / Image Upload	Capture river photo via mobile camera, webcam, or file upload
2	AI Detection	YOLOv8-nano	Detect and classify garbage into 4 types with bounding boxes
3	Risk Scoring	GDI Algorithm	Calculate weighted Garbage Density Index (0-100)
4	Classification	Pollution Levels	Assign Low / Moderate / High / Critical based on GDI
5	Storage	SQLite + FastAPI	Store results with GPS, timestamp, and image
6	Dashboard	Next.js + Chart.js	Visualize KPIs, trends, pie charts, scan history
7	Stakeholders	Web Interface	Field workers, monitoring officers, government authorities

5.2 Technology Stack

Component	Technology	Justification
Detection Model	YOLOv8-nano (Ultralytics)	Best speed/accuracy balance; proven in water-surface papers
Backend API	Python FastAPI	Async, modern, automatic OpenAPI documentation
Frontend	Next.js (React)	Server-side rendering, responsive, component-based
Database	SQLite (async)	Zero-configuration, serverless, lightweight
Training Platform	Google Colab (T4 GPU)	Free GPU access for model training
Dataset Management	Roboflow	Browser-based annotation, augmentation, YOLOv8 export
Charts	Chart.js + react-chartjs-2	Lightweight, responsive charting library

6. Dataset Preparation Strategy

6.1 Combined Dataset Approach

Following the established methodology of Cao et al. (2025), who combined 3,000 ORCA public images with 2,000 custom-collected images, AquaGuard uses a combined dataset strategy:

Source	Images	Context	Classes Covered	Purpose
Custom Sri Lankan dataset	~300	Sri Lankan rivers (Kelani, Maha Oya)	glass, metal, plastic, polythene	Local environmental context
Floating Garbage Detection (Roboflow)	3,431	Floating in water	plastic_bottle, polythene_bag, aluminum_can, chip_packet, cork_sheet	Volume and water-context variety
TOTAL (combined)	~3,731	All water/river context	All 4 target classes	Comprehensive training set

6.2 Class Mapping from Public Dataset

Public Dataset Class	Mapped To	Reasoning
plastic_bottle	plastic	Direct match - plastic container
polythene_bag	polythene	Direct match - polythene material
aluminum_can	metal	Aluminum is a metal
chip_packet	plastic	Chip packets are plastic packaging
cork_sheet	plastic	Cork sheets are synthetic/plastic material

6.3 Data Augmentation Applied

The following augmentations were applied via Roboflow to increase dataset variety:

Augmentation	Setting	Purpose
Horizontal flip	50% probability	Objects can appear from either direction
Vertical flip	50% probability	Water reflections create vertical symmetry
Rotation	+/-15 degrees	Camera angles vary at river locations
Brightness	+/-20%	Simulates sunny vs cloudy conditions
Mosaic	Enabled	Combines 4 images - improves small object detection

Mixup	15%	Blends images for better generalization

7. Methodology and Algorithm Selection

7.1 Why YOLOv8 Over YOLOv11

Factor	YOLOv8-nano	YOLOv11-nano	Winner
Stability	Very stable (2+ years)	Still developing	YOLOv8
Documentation	Extensive	Limited	YOLOv8
Research papers	100+ papers	~20 papers	YOLOv8
Community support	Huge community	Growing	YOLOv8
Small target detection	Good	Slightly better	YOLOv11
Parameters	3.01M	~2.6M	YOLOv11
Speed (FPS)	~100	~110	Similar

YOLOv8 was selected due to its proven stability, extensive documentation, and widespread validation in water-surface detection literature (Papers 1 and 4 both used YOLOv8 successfully). Future work could explore YOLOv11 for improved small-target detection capabilities.

7.2 GDI Calculation Methodology

Formula: GDI = Sum(confidence_i x danger_weight_i) / area_normalisation x scale

Danger Weights: polythene = 1.0 | plastic = 0.8 | glass = 0.7 | metal = 0.6

Pollution Levels: Low (0-25) | Moderate (25-50) | High (50-75) | Critical (75-100)

Example: An image with 3 polythene bags (conf 0.85) and 2 plastic bottles (conf 0.90):
GDI = ((0.85x1.0 + 0.85x1.0 + 0.85x1.0) + (0.90x0.8 + 0.90x0.8)) / 1.0 x 20 = 79.8 -> CRITICAL

7.3 Training Configuration

Parameter	Value	Reasoning
Model	YOLOv8-nano	Smallest variant - suitable for real-time inference
Image size	640 x 640	Standard YOLO input resolution
Epochs	100-150	More epochs for small dataset
Batch size	16	Balanced GPU memory usage
Optimizer	Adam	Adaptive learning rate - good for small datasets
Learning rate	0.0005 (fine-tune)	Lower rate preserves pre-learned features

Patience	25-30	Early stopping if no improvement
Augmentation	Mosaic + Mixup + Erasing	Maximum training variety

8. Implementation Plan — 1-Week Sprint

Day	Task	Deliverable
Day 1	Capture 200+ more photos (focus on glass, metal, polythene) Visit 2-3 rivers with iPhone	Raw photos on computer
Day 2	Upload to Roboflow, annotate all new images Label every object with bounding boxes	Annotated dataset in Roboflow
Day 3	Merge custom + public datasets in Roboflow Apply augmentation, export YOLOv8 format Upload to Google Drive	Combined dataset ZIP on Google Drive
Day 4	Train model on Google Colab (100 epochs) Download best.pt, graphs, metrics Replace model in backend	Trained model with accuracy results
Day 5	Test complete system end-to-end Upload 15-20 images through frontend Fix any bugs	Fully working system with test data
Day 6	Prepare presentation slides Practice demo flow Prepare backup images for live demo	Presentation ready
Day 7	Final testing and rehearsal Buffer day for any remaining issues	Demo-ready system

9. Evaluation Metrics

Metric	Formula	What It Measures	Target
Precision	TP / (TP + FP)	Of all detections, how many were correct?	>= 0.80
Recall	TP / (TP + FN)	Of all real objects, how many were found?	>= 0.75
F1-Score	2 x (P x R) / (P + R)	Balanced measure of Precision and Recall	>= 0.75
mAP@50	Mean AP at IoU 0.5	Overall accuracy across all classes	>= 0.75
mAP@50-95	Mean AP at IoU 0.5-0.95	Strict accuracy with precise boxes	>= 0.50
Inference time	ms per image	Speed of detection	<= 50ms

10. Presentation Guide for Research Panel

10.1 Recommended Demo Flow (15 minutes)

Time	What to Show	What to Say
0-2 min	Problem statement + research gap	"Existing papers focus only on detection. None provide actionable pollution management."
2-4 min	System architecture diagram	"Our 7-layer pipeline goes from camera input to cleanup prioritization."
4-6 min	Live demo: upload garbage image	"Watch the system detect, classify, calculate GDI, and assign pollution level."
6-8 min	Upload 3 more images with different locations	"Each scan is GPS-tagged. The dashboard tracks pollution across locations."
8-10 min	Show dashboard: KPIs, charts, history	"The pie chart shows 70% of pollution is polythene - this informs policy decisions."
10-12 min	Show training results + model accuracy	"The model achieves XX% mAP@50 on our combined dataset of 3,731 images."
12-15 min	Novelty explanation + future work	"We bridge the gap between detection and action. No existing paper does this."

10.2 Anticipated Panel Questions and Answers

Panel Question	Your Answer
Why YOLOv8, not YOLOv11?	"YOLOv8 was selected for its proven stability and extensive validation in water-surface detection. Papers by Cao (2025) and Dong (2026) both validated YOLOv8 for this domain. YOLOv11 exploration is planned as future work."
Why not create all your own data?	"Following Cao et al. (2025) methodology, we combined custom Sri Lankan images with a public dataset. Our custom images provide local context while the public dataset provides volume. This is standard research practice."
What is different from existing work?	"All 5 reviewed papers stop at detection. Our system adds weighted GDI scoring, location tracking, temporal analysis, and a complete web-based monitoring platform. We transform passive detection into active pollution management."
How accurate is your model?	"The model achieves [your mAP]% overall, with plastic being the strongest class. Accuracy will improve as we expand the custom Sri Lankan dataset."
Why only 4 classes?	"These 4 classes represent the most common and environmentally harmful floating waste types in Sri Lankan rivers: plastic bottles, polythene bags, glass bottles, and metal cans. This focused approach ensures higher per-class accuracy."

Final Statement for the Panel:

"Our contribution is not a new detection algorithm - it is the complete pipeline from detection to actionable intelligence. Existing research improves mAP scores but provides no mechanism for environmental authorities to actually manage river pollution. AquaGuard bridges this gap by introducing the Garbage Density Index, location-based pollution tracking, and the first visual garbage detection system designed for Sri Lankan river environments. This transforms passive detection into an active environmental management tool."